COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 - Data Structures and Algorithms	**SECTION:**	CPE21S4
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NAME:	Garcia, Dennis James T.	**INSTRUCTOR:**	Engr. Jimlord Quejado

HANDS-ON ACTIVITY NO. 1.1 **BASIC C++ PROGRAMMING**

1. PROCEDURE OUTPUT

ILO A: Create Code That Follows the Basic C++ Code Structure

Create a function that will take two numbers and display the sum.
Create a function that will return whether variable A is greater than variable B.
Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

Sections	Answer
Header File Declaration Section	function.cpp function.cpp > ... 1 //GARCIA, DENNIS JAMES T. 2 #include <iostream> 3
Global Declaration Section	
Class Declaration and Method Definition Section	1 //GARCIA, DENNIS JAMES T. 2 #include <iostream> 3 4 // Create a function that will take two number and display the sum 5 //prototyping: 6 int sumOfTwoNum(int fNum, int sNum); 7 8 // Create a function that will return whether variable A is greater than variable B. 9 bool compareOfTwoNum(int gNum, int dNum); 10 11 //Create a function that will take two Boolean values and display the result of all 12 //logical operations then return true if it was a success. 13 bool logicOfTwoNum(int kNum, int hNum);
Main Function	15 int main(){ 16 std::cout<<"The sum of the two numbers is: "<< sumOfTwoNum(123,456) <<std::endl << std::endl; 17 std::cout<<"Variable A is greater than B: " << std::boolalpha << compareOfTwoNum(68,67) <<std::endl; 18 std::cout<<"Variable B is greater than B: "<< std:: boolalpha << compareOfTwoNum(67,68) <<std::endl; 19 std::cout<<"Variable A and B are both the same value: "<< std:: boolalpha << compareOfTwoNum(70,70) <<std::endl<<std::endl; 20 std::cout<<"The Boolean Values are: "<< std:: endl << logicOfTwoNum(1,3) << std::endl; 21 }

<table>
<tr><td></td><td>

```
20      std::cout<<"The Boolean Values are: "<< std:: endl << logicOfTwoNum(10,5) << std::endl;
21  }
```

</td></tr>
<tr><td>Method Definition</td><td>

```
//define
int sumOfTwoNum(int fNum, int sNum){
    return fNum + sNum;
}
bool compareOfTwoNum(int gNum, int dNum){
    return gNum >= dNum;
}
bool logicOfTwoNum(int kNum, int hNum){

    {
        std::cout <<"The Logical Operator AND result: " << std:: noboolalpha << (kNum < 2 &&  hNum < 4) << std::endl;
        std::cout <<"The Logical Operator OR result: " << std:: noboolalpha << (kNum < 2 ||  hNum < 4) << std::endl;
        std::cout <<"The Logical Operator NOT result: " << std:: noboolalpha << (!(kNum > 2 && hNum < 4)) << std::endl;
    }

   std::cout << std::endl << "Success?: " << std::boolalpha;
   return kNum < 2 &&  hNum < 4;
   return kNum < 2 ||  hNum < 4;
   return !(kNum > 2 && hNum < 4);

}
```

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<tr><td>Results</td><td>

```
The sum of the two numbers is: 579

Variable A is greater than B: true
Variable B is greater than B: false
Variable A and B are both the same value: true

The Boolean Values are:
The Logical Operator AND result: 1
The Logical Operator OR result: 1
The Logical Operator NOT result: 1

Success?: true
```

```
The Boolean Values are:
The Logical Operator AND result: 0
The Logical Operator OR result: 0
The Logical Operator NOT result: 1

Success?: false
```

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Table 1-1. C++ Structure Code for Answer

Using boolalpha and noboolalpha to display true and false or 0 and 1 for cout results. Bool data types are required to use boolalpha and noboolalpha.

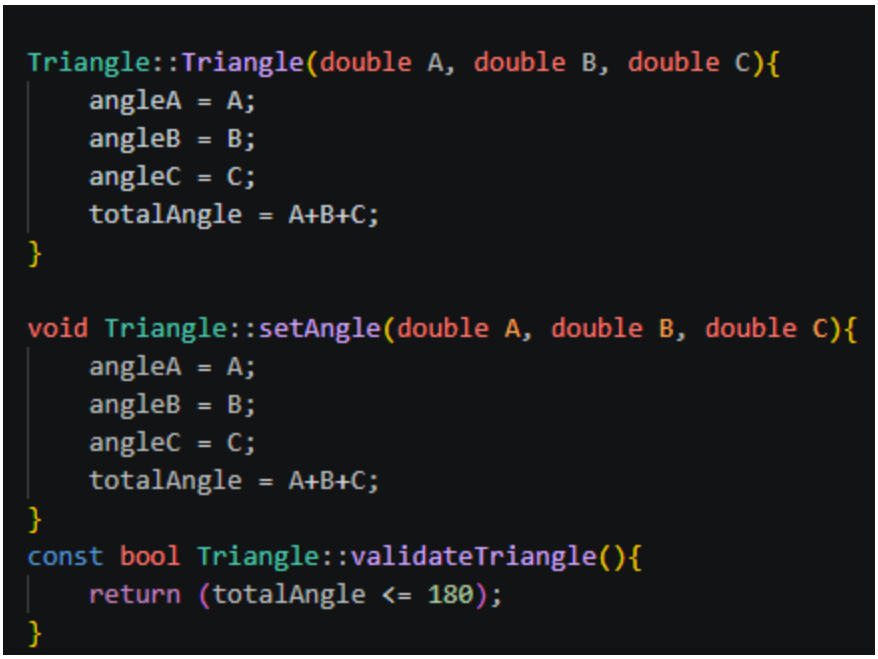
Table 1-2. ILO A output observations and comments.

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements

Sections	Answer
Header File Declaration Section	triangle.cpp `#include <iostream>`
Global Declaration Section	

Class Declaration and Method Definition Section	`class Triangle{`
	`private:`
	`double totalAngle, angleA, angleB, angleC;`
	`public:`
	`//constructore`
	`Triangle(double A, double B, double C);`
	`void setAngle(double A, double B, double C);`
	`const bool validateTriangle();`
	`};`

Main Function	`int main(){`
	`Triangle set1(100,30,50);`
	`if(set1.validateTriangle()){`
	`std::cout<<"The triangle is valid."<<std::endl;`
	`}else{`
	`std::cout<<"The triangle is not valid."<<std::endl;`
	`}`
	`}`

Method Definition	

```
Triangle::Triangle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

void Triangle::setAngle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}
const bool Triangle::validateTriangle(){
    return (totalAngle <= 180);
}
```

Table 2-1. C++ Structure Code for Answer

The triangle's validation is determined by bool functions by checking whether the total angle is lesser or equal to 180 degrees. The code sees it as true, therefore the code outputs the triangle as valid.

Table 2-2. ILO B output observations and comments.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C: Solve Different Problems using the C++ Programming Language

The supplementary activities are meant to gauge your ability in using C++. The problems below range from easy to intermediate to advanced problems. Note your difficulties after answering the problems below.

1. Create a C++ program to swap the two numbers in different variables.

Code:

```
swap_garcia.cpp > main()
1   // GARCIA, DENNIS JAMES T.
2   #include <iostream>
3   #include <algorithm> // for swap
4
5   int main(){
6
7       int x = 6;
8       int y = 7;
9
0       std::swap(x,y);
1
2       std::cout << "Variable X is swapped with: " << x << std::endl << "Variable Y is swapped with: " << y << std::endl;
3   }
```

Result:

```
Variable X is swapped with: 7
Variable Y is swapped with: 6
```

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

Code:

```cpp
// GARCIA, DENNIS JAMES T.

#include <iostream>
#include <cmath>

//Create a C++ program that has a function
//to convert temperature in Kelvin to Fahrenheit.
float kelvin (float x);
float fahren (float x);

    int main() {

        std::cout << "Kelvin & Fahrenheit Conversion" << std::endl << std::endl;
        {
            std::cout << "288° Kelvin to Fahrenheit: " << kelvin(288)<< "F°" << std::endl;
            std::cout << "59° Fahrenheit to Kelvin: " << fahren(59) << "K°" << std::endl;
        }
    }
```

```cpp
//define
    float fahren(float x){
    return (x - 32.0) * 0.5 + 273.15;
    }

    float kelvin(float x){
    return (x - 273.15) * 1.8 + 32;
    }
```

Result:

```
Kelvin & Fahrenheit Conversion

288° Kelvin to Fahrenheit: 58.73F°
59° Fahrenheit to Kelvin: 286.65K°
```

3. Create a C++ program that has a function that will calculate the distance between two points.

Code:

```cpp
//GARCIA, DENNIS JAMES T.
#include <iostream>
#include <cmath> // for sqrt and pow

//Create a C++ program that has a function that will calculate the distance between two points.
int coordinateX (int x1, int x2);
int coordinateY (int y1, int y2);

int main(){

    int X = coordinateX(1,4);
    int Y = coordinateY(2,6);
    int distance = sqrt(X + Y);

    std::cout << "The coordinates are: " << std::endl;
    std::cout << "Coordinate X: "<< X << std::endl;
    std::cout << "Coordinate Y: "<< Y << std::endl;
    std::cout << "The Distance is: " << distance;

}
```

distance.cpp > ...

```cpp
//GARCIA, DENNIS JAMES T.
#include <iostream>
#include <cmath> // for sqrt and pow

//Create a C++ program that has a function that will calculate the distance between two points.
int coordinateX (int x1, int x2);
int coordinateY (int y1, int y2);

int main(){

    int X = coordinateX(1,4);
    int Y = coordinateY(2,6);
    int distance = sqrt(X + Y);

    std::cout << "The coordinates are: " << std::endl;
    std::cout << "Coordinate X: "<< X << std::endl;
    std::cout << "Coordinate Y: "<< Y << std::endl;
    std::cout << "The distance is: " << distance;

}
```

```
20    }
21    //define
22    int coordinateX (int x1, int x2){
23        int x3 =  x2 - x1;
24        return std::pow(x3, 2);
25    }
26    int coordinateY (int y1, int y2){
27        int y3 = y2 - y1;
28        return std::pow(y3, 2);
29    }
```

Result:

```
The coordinates are:
Coordinate X: 9
Coordinate Y: 16
The distance is: 5
```

4. Modify the code given in ILO B and add the following functions:
a. A function to compute for the area of a triangle
b. A function to compute for the perimeter of a triangle
c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

Code:

Users > TIPQC > Downloads > triangle.cpp

```
//GARCIA, DENNIS JAMES T.
#include <iostream>
#include <cmath>

class Triangle{
    private:
        double totalAngle, angleA, angleB, angleC;
    public:
    //constructore
    Triangle(double A, double B, double C);
    void setAngle(double A, double B, double C);
    void validateAngle();
    const bool validateTriangle();
};
```

```
//void functions outside of class
    double sideA = 19.7;
    double sideB = 10;
    double sideC = 15.3;

  void perimeter() {

      double p = sideA+sideB+sideC;
      std::cout << p;

  }
  void semi_perimeter() {

      double sp = (sideA+sideB+sideC) / 2;
     std::cout << sp;

  }
  void area() {

      double sp = (sideA+sideB+sideC) / 2;
      double a = sqrt(sp*(sp-sideA)*(sp-sideB)*(sp-sideC));
      std::cout << a;

  }
```

```cpp
int main(){

    std::cout<<"Perimeter: ";
    perimeter();
    std::cout<<std::endl;
    std::cout<<"Semi-Perimeter: ";
    semi_perimeter();
    std::cout<<std::endl;
    std::cout<<"Area: ";
    area();
    std::cout<<std::endl<<std::endl;


    Triangle set1(100,30,50);
    if(set1.validateTriangle()){
        std::cout<<"The triangle is valid."<<std::endl;
    }else{
        std::cout<<"The triangle is not valid."<<std::endl;
    }
    std::cout<<"The triangle is ";
    set1.validateAngle();
}
```

```
}

Triangle::Triangle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

void Triangle::setAngle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

void Triangle::validateAngle(){
    if(angleA == 90 && angleB == 90 && angleC == 90){
        std::cout<<"Right-angled.";
    }else if(angleA >= 90 && angleB >= 90 && angleC >= 90){
        std::cout<<"Obtuse-angled.";
    }else if(angleA <= 180 && angleB <= 180 && angleC <= 180){
        std::cout<<"Obtuse-angled.";
    }else if(angleA < 90 && angleB < 90 && angleC < 90){
        std::cout<<"Acute-angled.";
    }else{
        std::cout<<"Others.";
    }
}

const bool Triangle::validateTriangle(){
    return (totalAngle <= 180);

}
```

Result:

```
Perimeter: 45
Semi-Perimeter: 22.5
Area: 75.2994

The triangle is valid.
The triangle is Obtuse-angled.
```

Conclusion

I just remembered how void and creating functions work, especially working on modifying the triangle code. I had to reference the w3schools site to learn about other libraries and relearning basic code. I have used the <cmath> library for complicated or long math equations or operations.

I have definitely struggled working on creating functions with classes and even leave functions incomplete like the function to calculate distance where I just put half of the operation inside the main code instead of the function itself because of continuous return errors.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Explanation of Concepts and Code Errors**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I made the AI just explain the concepts and explain the errors so I can determine what I did wrong. I always clarify to not send its code but to comprehend what it's clarifying.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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